

DS	Session	Queue	Question Information	Level 1	Challenge 60
Question description					
Umesh is an DS expert training youngsters struggling in DS to make them better.					
Umesh usually gives interesting problems to the youngsters to make them love the DS.					
One such day Umesh provided to the youngsters to solve the task such that, Reverse a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first.					
Youngsters were lacking the idea to solve the problem.					
Being an exciting youngster can you solve it?					
Function Description					
• INITIALIZE FRONT AND REAR = -1 • CALL ENQUEUE FUNCTION WHILE(I<GIVEN SIZE OF QUEUE) • CALL REVERSE • MAKE A FOR LOOP(I=FRONT,J=REAR;I<J;I++;J++) • SWAP(FRONT,REAR)					
Constraints:					
0<size <100					
0<data<1000					
Input format:					
First line indicates the size of the queue					
Second line indicates the elements of the queue.					
Output Format:					
first line indicates the enqueue of each elements					
last line indicates the reversed queue elements.					

Test Case 1

INPUT (STDIN)

5
5 6 3 1 2

EXPECTED OUTPUT

Queue:5 6 3 1 2
Reversed Queue:2 1 3 6 5

Test Case 2

INPUT (STDIN)

8
1 8 6 4 9 5 3 5

EXPECTED OUTPUT

Queue:1 8 6 4 9 5 3 5
Reversed Queue:5 3 5 9 4 6 8 1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

`void enqueue(int data,int l)`

Test Case 2

KEYWORD

`void reverse()`

Test Case 3

KEYWORD

`enqueue(t,n);`

Test Case 4

KEYWORD

`reverse();`

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

57

Queue10.c +

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1;
7 int rear = -1;
8
9 void enqueue(int data, int n) {
10     if (rear == MAX - 1) return;
11     if (front == -1) front = 0;
12     queue[++rear] = data;
13 }
14
15 void reverse() {
16     int i = front, j = rear;
17     while (i < j) {
18         int temp = queue[i];
19         queue[i] = queue[j];
20         queue[j] = temp;
21         i++;
22         j--;
23     }
24 }
25
```

```
26 int main() {  
27     int n;  
28     if (scanf("%d", &n) != 1) return 0;  
29  
30     for (int i = 0; i < n; i++) {  
31         int t;  
32         scanf("%d", &t);  
33         enqueue(t, n);  
34     }  
35  
36     printf("Queue:");  
37     for (int i = front; i <= rear; i++) {
```

```
38     printf("%d%c", queue[i], (i == rear) ? '\n' : ' ');
39 }
40
41 reverse();
42
43 printf("Reversed Queue:");
44 for (int i = front; i <= rear; i++) {
45     printf("%d%c", queue[i], (i == rear) ? '\n' : ' ');
46 }
47 return 0;
48}
49
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5

5 6 3 1 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue10.exe

Queue:5 6 3 1 2

Reversed Queue:2 1 3 6 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

8

1 8 6 4 9 5 3 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue10.exe

Queue:1 8 6 4 9 5 3 5

Reversed Queue:5 3 5 9 4 6 8 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>